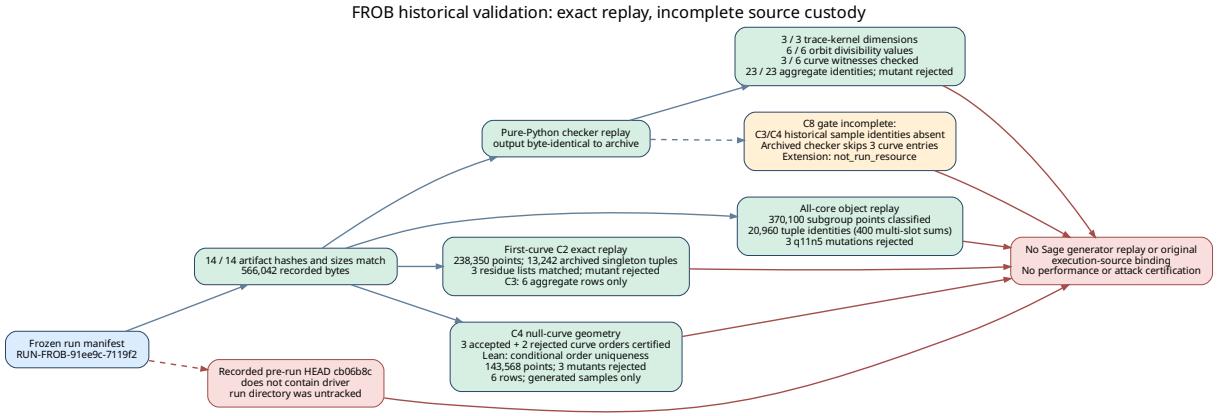
Historical FROB checker replay and source custody

Validation audit, 22 October 2026

Scope. Frozen `RUN-FROB-91ee9c-7119f2` at tree `e83bb2f14414c9669360816aeee5876dfcfd4384`; copied run artifacts, no new measurement or research verdict.

Check	Result	Limit
Archived artifact hashes and sizes	14/14 match; 566,042 bytes	Storage integrity only
Checker output	Byte-identical replay	Same archived checker source
Trace-kernel dimension; orbit divisibility	3/3; 6/6	Recorded core cells
Curve point/order/Frobenius witnesses	3/6 checked	Other three skipped for size
Aggregate rational identities	23/23; mutant rejected	Counts treated as inputs
New all-core object-arm replay	370,100 points; 20,960 tuples	400 multi-slot sums; 3 q11 mutants
First-curve C2 replay	238,350 points; 13,242 tuples	3 exact residue lists; singleton only
C3 aggregate replay	6 rows; 26,620 generated tuples	Archived sampled lists absent
C4 null geometry and Lean arithmetic	5 conditional exact orders; 143,568 points	Hasse/point-order premises; samples absent
Remaining C8 work	Open	C3/C4 historical sample identities
Extension cell	`not_run_resource`	540-second historical checkpoint



Source custody. The manifest's pre-run HEAD `cb06b8c20c2d6559041c45c4a4b29dd7f132fdf5` lacks the driver; the run directory was untracked. Hash-pinned command and environment files do not bind that HEAD to the executed source.

Boundary. The archived checker recomputes no tuple sums. New replays cover object arms, C2 lists and C4 geometry. Lean proves order uniqueness conditional on the interval and point-order inputs; those inputs and the historical C3/C4 sample identities are not formalized. Source custody, C8 and the success gate remain unestablished.

Receipts: Exact replay and Lean verification JSON files are linked in `REPORT.md`.